Homework 1: Image Enhancement

Computer Vision – Fall 2026

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Repository: https://github.com/Shihab7u/Image-Enhancement
(code, images, results, metrics.csv, README)

1 Objective and data

The task is to create and evaluate a simple classical (non-deep learning) pipeline for enhancing colour images with low contrast, non-uniform lighting, and noise, and to evaluate the trade-offs between the visibility of details, noise and naturalness of the image.

Images. I will use five images from my own smartphone photographs of outdoor scenes (no synthetic images). **Source and licence:** own photographs, taken by the author and redistributed by the author in the repository **(data/raw/)**. To save time, each 1932×2576 image is downscaled to 720×960 px using area interpolation **(src/prepare_data.py)**; PNGs are the originals.

Image	Condition exhibited
forest	dark, back-lit forest canopy; sun in frame with lens flare; shadowed trunks
waterfall_haze	haze and back-light; dark vegetation behind the falls; sun rays
river_boat	atmospheric haze, low contrast on the distant hillside
waterfall_canyon	harsh sunlight with deep shadows in the gorge (non-uniform illumination)
hut	large bright sky vs. darker hillside and ground; haze

Synthetic degradation. For reproducible evaluation each original I is degraded as

$$I_d = \mathrm{clip}_{[0,255]}(\mathrm{round}(\mu + c\,(I - \mu) + n)), \qquad c = 0.5, \quad n \sim \mathrm{N}(0, 6^2),$$

where μ is the global mean of I (image brightness is preserved, dynamic range is halved) and n is independently and identically distributed (i.i.d.) per pixel and channel, using **numpy.random.default_rng(42+k)** for the k-th image. Both are saved. The original images only provide a basis for comparing these artificial corruptions; in practical applications, a clean image usually will not be available. Since the contraction is performed in RGB space relative to a grey value, it also decreases color saturation (e.g. mean Lab chroma of hut: 25.0 → 14.9, 8-bit OpenCV units).

2 Methods and parameters

All methods start from the degraded 8-bit image and use OpenCV **[4]** and NumPy **(src/enhance.py)**. Background is described in **[5].** All techniques have the same data, degradations, and evaluation conditions.

1. **Baseline**: the degraded image, as-is.
2. **Linear contrast stretching**: the [p1, p99] percentile range is mapped linearly to [0, 255] (1% saturated at both ends, resistant to outliers). Variants include the application on luminance L∗ within the CIE-Lab colour space, while leaving the a∗, b∗ components untouched, thus keeping hue and not creating any color cast. To compare we also apply contrast stretching to each channel (B, G, R independently), where every channel will have its own gain/offset, equivalent to an automatic white balance.
3. **Histogram equalisation**: Histogram equalization: global cv2.equalizeHist on L∗, the chroma component is left as is. By applying histogram equalization to L∗ instead of R, G, B, we avoid any hue shift, but since L∗ varies a lot whereas a∗ and b∗ remain constant, saturation changes.
4. **Fourth method – NLM denoising + luminance stretch** (motivated combination). Stretching magnifies the noise to the same extent that it does the signal (which is 1/c ≈ 2× for us), so we denoise first, using non-local means **[1], cv2.fastNlMeansDenoisingColored** (h=hcolor=4, template 7, search 21), followed by method 2 applied to L*. Removing noise has the inevitable side-effect of smoothing textures.

Metrics (src/metrics.py), on the degraded set against the original: **PSNR** (own implementation, peak 255, all pixels and channels), **SSIM [2]** (scikit-image **[3]**, Gaussian window σ = 1.5, mean over channels) and **MAE** in grey levels.

3 Results

Figure 1 shows the side-by-side comparisons for each image and algorithm using the same fixed [0, 255] dis-play range; Figure 2 shows 1:1 crops that allow for seeing the noise and fine details; the metrics are shown in Table 1 (also **results/metrics.csv**).

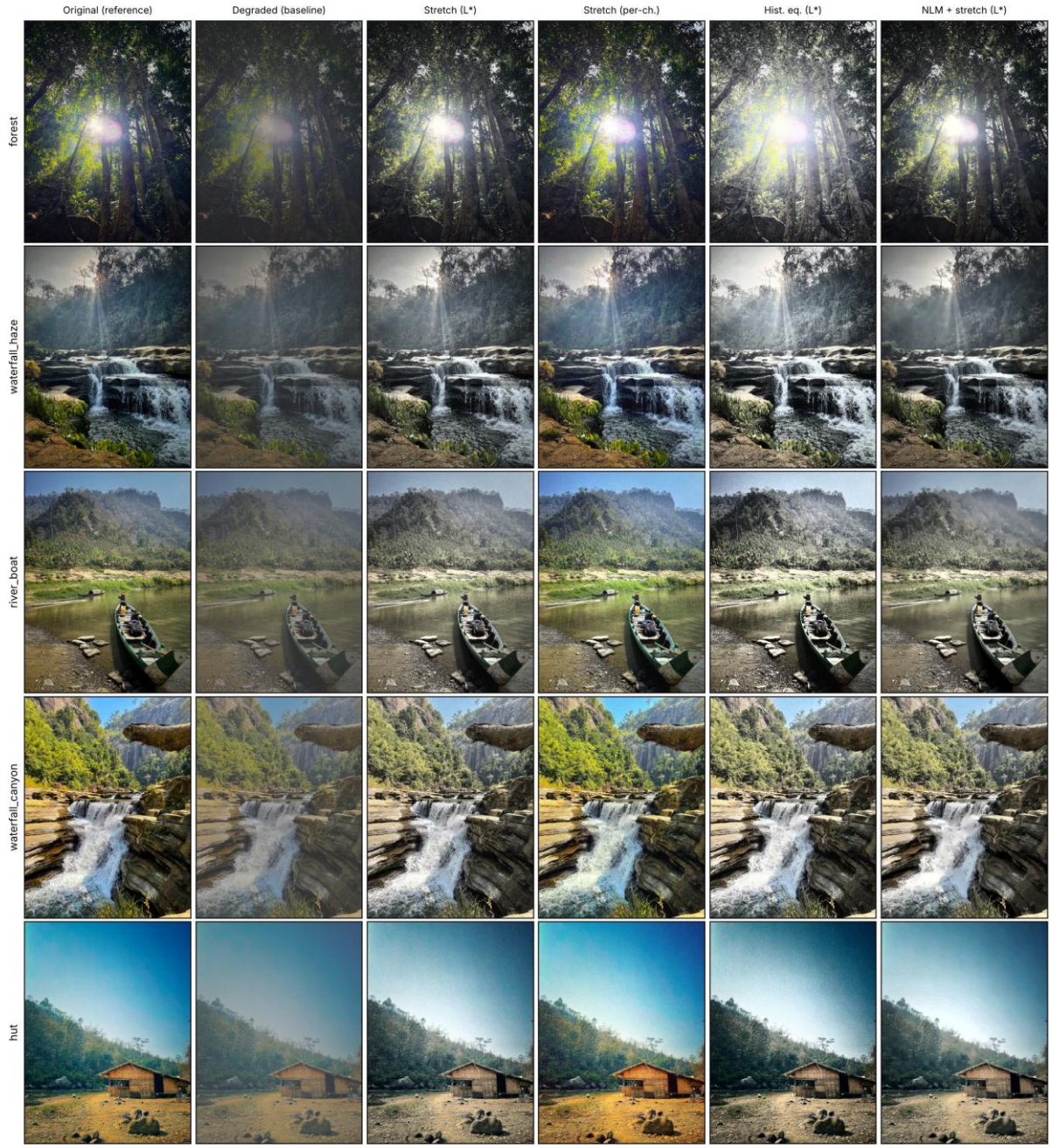


Figure 1: Original (reference), degraded input (= baseline) and the four enhancement results for each image.

4 Discussion

General trends. L* stretching provides better PSNR compared to the baseline method in four out of five cases and NLM+stretch for all five cases, while the histogram equalization has the lowest PSNR in four out of five cases. The HE does not work well due to the forced flat L* histogram, where the dark forest (L* mean = 67/255) will be boosted to mid-grey, while the large bright area of sun/flare becomes an extremely large white spot (Figure 2, row 1). All stretching techniques increase the noise in a flat area of the hut sky the high frequency L* standard deviation increases from 4.7 (degraded) to 11.3 after L* stretching and to 16.1 after HE, while after NLM+stretching the deviation is 1.3 (original: 1.0). There are two systematic limitations that persist: (i) only L∗ methods are unable to recover the saturation lost due to the degradation process (average chroma of hut is 14.8 instead of 25.0 initially), which results in the wooden hut appearing beige-grey instead of

Table 1: Metrics with respect to the original (higher PSNR/SSIM and lower MAE are better). Bold: best per image.

Image	Method	PSNR [dB] □	SSIM □	MAE ↓
forest	Baseline (degraded)	19.69	0.667	20.36
	Linear stretch, L^*	22.67	0.708	13.86
	Linear stretch, per channel	22.25	0.622	15.97
	Hist. equalisation, L^*	11.29	0.412	61.21
	NLM + stretch, L^*	**22.97**	**0.790**	**12.13**
waterfall_haze	Baseline (degraded)	20.04	0.681	20.68
	Linear stretch, L^*	20.92	0.727	18.80
	Linear stretch, per channel	19.66	0.635	22.40
	Hist. equalisation, L^*	13.92	0.598	42.90
	NLM + stretch, L^*	**21.80**	**0.825**	**16.54**
river_boat	Baseline (degraded)	20.36	0.693	19.58
	Linear stretch, L^*	19.79	0.721	21.64
	Linear stretch, per channel	19.82	0.641	21.67
	Hist. equalisation, L^*	15.48	0.594	33.54
	NLM + stretch, L^*	**20.44**	**0.823**	**19.43**
waterfall_canyon	Baseline (degraded)	17.56	0.686	28.75
	Linear stretch, L^*	22.10	**0.826**	14.44
	Linear stretch, per channel	**25.04**	0.791	**11.40**
	Hist. equalisation, L^*	19.68	0.806	19.83
	NLM + stretch, L^*	21.09	0.825	15.96
hut	Baseline (degraded)	17.87	0.635	28.06
	Linear stretch, L^*	18.65	0.573	22.99
	Linear stretch, per channel	**21.36**	0.514	**17.92**
	Hist. equalisation, L^*	17.62	0.524	27.62
	NLM + stretch, L^*	18.42	**0.799**	23.52

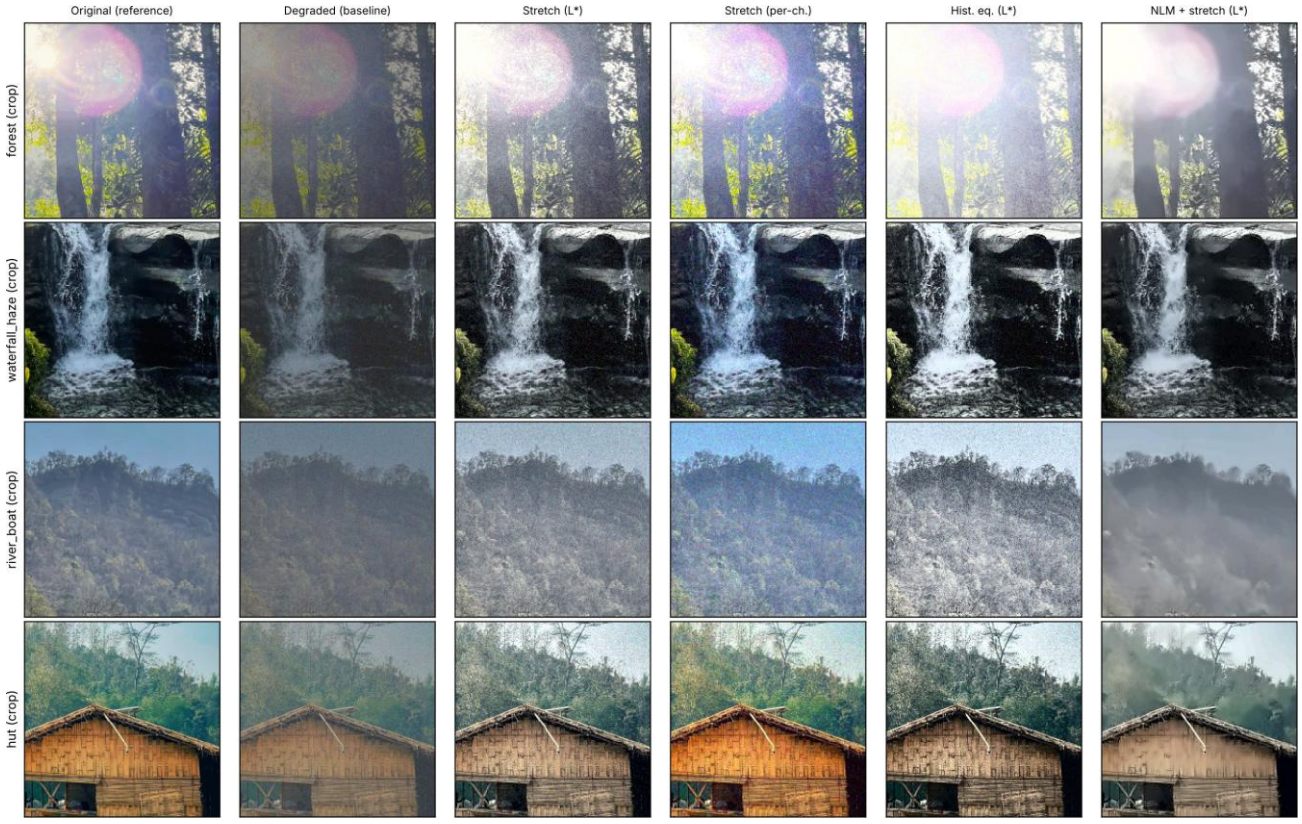


Figure 2: Zoomed crops (200×200 px, nearest-neighbour): flare region of forest, falls of waterfall_haze, distant hillside of river_boat, roof and hillside of hut.

orange; (ii) all of them are global, so they are not able to get rid of haze locally or even balance the sunlit and shadowed parts of waterfall_canyon..

Case 1 – better numbers, worse-looking image (hut, per-channel stretch). Per-channel stretching performs best for PSNR (21.36 dB) and MAE (17.92) on hut (and on waterfall_canyon, 25.04 dB), since the independent gains for each channel stretch the chroma, bringing the orange wood and blue sky closer to the reference. However, it has the lowest SSIM of all techniques on hut (0.514), and the crop demonstrates pronounced colored noise in the roof and sky areas. PSNR/MAE are determined mostly by the large low-frequency color error, while SSIM and the human eye are more susceptible to high-frequency noise. Per-channel gains amplify colors: the mean chroma on forest and **waterfall_haze** is 17.4, compared to 10 in the originals.

Case 2 – the "best" method erases detail (river_boat, NLM+stretch). NLM+stretch outperforms all other methods according to all three criteria on river_boat (PSNR 20.44, SSIM 0.823, MAE 19.43); however, the crop of the distant hillside proves that tree crowns have been smoothed into painter-like areas. The texture is already very poor because of the haze, and therefore, the contrast reduction leads to an almost equal amplitude of the signal and noise; both are removed by NLM. L∗ stretching retains the visibility of the trees despite the grain, yet has a worse PSNR compared to the original image

(19.79 vs. 20.36 dB), since the 1–99% stretch corresponds to the whole [0, 255] range, while the original image has values in the range [2, 218] (p1–p99 of L∗). The reference-based metrics evaluate the quality of the restoration, not enhancement.

Qualitative analysis. forest – preferred: NLM+stretch (best for all criteria). The trunks, foliage, and light rays reappear, and the grain in the dark areas of the canopy is cleaned up. Artifact: the flare zone becomes slightly mottled with noticeable tonal steps. HE shows the most detail in the leaves in the shadow areas but floods the entire scene with an unnatural yellowish white around the sun, lifting the shadows to an unnatural grey. **waterfall_haze**, preferred: NLM+stretch (best for all criteria). The water texture and rocks retain their details since they are high-contrast features and NLM preserves them; the haze and sun rays are preserved too. HE creates a milky-white haze and introduces noticeable noise. Hut - no method gives

perfect results: L∗ stretching brings back the contrast of the hillside, but it destroys the colour of the wood; per-channel stretching preserves the colour but introduces colour noise; NLM+stretch provides a clean sky but makes the foliage of the hillside appear in water-colour style (see Fig. 2, last row). I prefer L∗ stretching in this case for its natural look and preserved detail.

Real images, no reference. When applied directly to the raw images (Fig. 3, **results/images/** files *_REAL_*), that is what happens. L∗ compression has only slight effects, because the photos already cover almost the whole range. HE has more noticeable effects, though; it helps to show the texture in the forest of **waterfall_haze** and on the **river_boat** hill side, however it brightens too much the **forest** (sun glare spreads across the canopy).

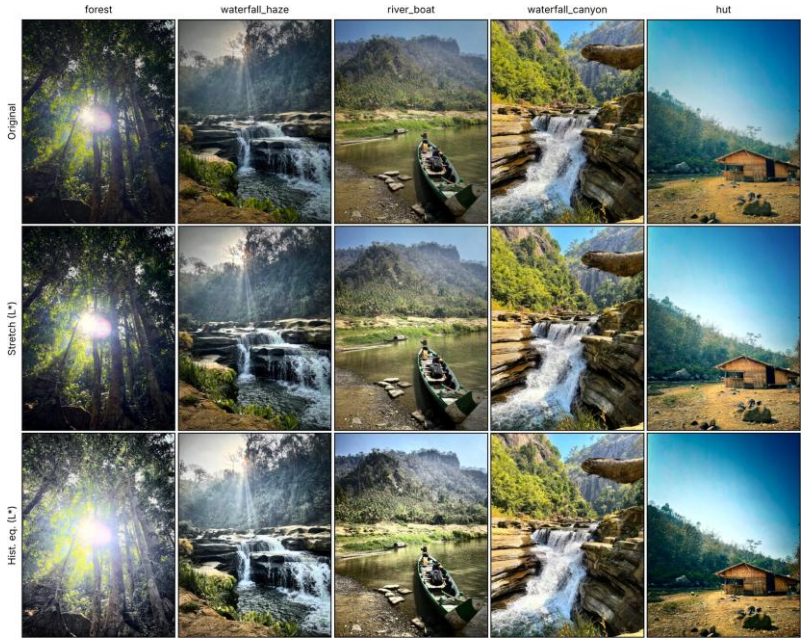


Figure 3: Methods applied to the original photographs (no reference available).

Limitations. Degradation is global with simple Gaussian noise (real low light noise is signal dependent); one parameter setting per method is used; the images have been resampled; and all measures are full-reference fidelity measures which do not measure visual pleasantness. We thus do not select a method based on a single measure: our default is NLM + luminance stretch for noisy images, and luminance stretch alone for fine textures (distant foliage).

References

[1] A. Buades, B. Coll, J.-M. Morel. A non-local algorithm for image denoising. *CVPR*, 2005.

[2] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Processing*, 13(4), 2004.

[3] S. van der Walt et al. scikit-image: image processing in Python. *PeerJ* 2:e453, 2014.

[4] G. Bradski. The OpenCV Library. *Dr. Dobb's Journal of Software Tools*, 2000.

[5] R. C. Gonzalez, R. E. Woods. *Digital Image Processing*, 4th ed. Pearson, 2018.